

FX SPATIAL Technical Manual

Delay, reverb, tail, and spatial environment

Create audible space, movement, and controlled tails on parallel spatial lanes.

Included products: NTRSTLLR, Cathedral, SacredVerb

Routing authority: Parallel spatial route: NTRSTLLR -> Cathedral -> SacredVerb.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

NTRSTLLR

Audible interstellar delay and tail engine with travel, warp, orbit, portal, delay, air, haze, width, diffusion, axis, and Nebula/Wormhole modes.

Declared product: NTRSTLLR | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
trumpet_l	Micro Peak	Float
trumpet_c	Meso Density	Float
trumpet_r	Macro Level	Float
makeup_gain	Makeup Gain	Float
travel	Travel	Float
warp	Warp	Float
orbit	Orbit	Float
portal	Portal	Float
time_delay	Time/Delay	Float
rubber	Rubber	Float
air	Air	Float
haze	Haze	Float
width	Width	Float
diffusion	Diffusion	Float
axis	Axis	Float
mix	Mix	Float
output	Output	Float
wormhole_mode	Wormhole Mode	Bool
nebula_mode	Nebula Mode	Bool
phi_grid	Phi Grid	Bool
power	Power	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/NTRSTLLR_BG_20260715.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT / OUTPUT	Weighted peak-plus-RMS level	Use for level matching around the spatial return.

Display	What it measures	How to read it
TONE RESPONSE	Current low-to-high response shape	Shows tonal contour, not a mastering EQ recommendation.
WORMHOLE	Delay/warp-path energy	Shows active wormhole return strength.
NEBULA	Portal/diffusion-path energy	Shows active nebula tail strength.
SINGULARITY	Axis and pull concentration	Higher values indicate stronger inward focus.
PHI FIELD OBSERVATION	Combined macro-control activity	A spatial-state summary rather than loudness.
PHI ECHO / DELAY FIELD	Delay taps, orbit, and return motion	Use to understand tail movement and density.

Cathedral

Seven-river chamber environment with arch, convergence, 3-6-9 nodes, protected I/O, and parallel mix.

Declared product: Cathedral | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
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Parameter surface

ID	Display name	Type
arch	Arch	Float
rivers	Seven Rivers	Float
nodes	3-6-9 Nodes	Float
convergence	Convergence	Float
input	Input	Float
mix	Mix	Float
output	Output	Float
channel_mode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Artwork authority

Assets/Images/Cathedral.png

Meter and visualizer semantics

Display	What it measures	How to read it
ENERGY	Current chamber RMS in dB	Shows how strongly the room is being excited.
DIFFUSION	3-6-9 node density	Higher values produce a denser, smoother tail.
WIDTH	Node density multiplied by wet mix	Shows audible spatial spread, not dry-source width.
RT60	Estimated mid-band decay time	Seconds required for the modeled room to decay by 60 dB.
ROOM RESONANCE	Modeled low/mid/high response	Use to identify room emphasis before printing.

SacredVerb

Sacred spatial reverb with pre-delay, chamber size, damping, rate, depth, Neter blend, resonance, chamber and pattern modes, and protected presets.

Declared product: SacredVerb | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
PRE_DELAY	PRE DELAY	Float
ROOM_SIZE	CHAMBER SIZE	Float
DAMPING	DAMPING	Float
REVERB_MIX	WET MIX	Float
MOD_RATE	RATE	Float
MOD_DEPTH	DEPTH	Float
NETER_BLEND	NETER BLEND	Float
RESONANCE	RESONANCE	Float
INPUT	INPUT	Float
OUTPUT	OUTPUT	Float
CHAMBER	Chamber	Int
PATTERN_MODE	Pattern Mode	Int
FLOWER	Flower of Life	Bool
ECO_MODE	Eco Mode	Bool
BYPASS	Bypass	Bool
gainmatch	Gain Match	Bool
TRIG_TAIL	Trigger Tail	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
TRIG_	TRIG_	Referenced
EN_	EN_	Referenced

Artwork authority

Assets/Images/SacredVerb.png

Meter and visualizer semantics

Display	What it measures	How to read it
Seven Hathor meters	Energy at enabled Solfeggio nodes	Each column shows its resonant band activity and peak hold.
Output meter	Current reverb-output peak	Use to keep the return controlled and level matched.
3-6-9 modulator	Active modulation/pattern state	Shows motion settings; it is not a loudness meter.